

PTJC: Anomaly Cancellation in the Standard Model

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Tong's Standard Model + Gauge Theory Notes, Weinberg Chp. 22

1 Brief Review

What have we done so far? Murali introduced anomalies, which occur when quantum effects cause conserved quantities/currents to no longer be conserved. We saw that in the Schwinger Model on a circle, the current $j_5^\mu = \bar{\Psi}\gamma^\mu\gamma^5\Psi$ was no longer conserved, since by gauging A_1 from 0 to $2\pi/L$, we raise a LH fermion out of the Dirac sea and drop a RH fermion into it, leading to non-conservation of axial charge Q_5 .

We then looked at two ways to calculate the chiral anomaly, i.e. the non-conservation of axial current in a more general theory of massless fermions. Via showed us how to derive this anomaly from summing triangle diagrams, while Cameron showed us how to derive it from the transformation of the path integral measure under an axial rotation of our fermion fields.

2 Gauge Anomaly Cancellation

We don't want gauge anomalies, because they break gauge invariance. We don't want to apply a gauge transformation and not get back to where we started. Tong comments that this has to do with the creation of negative norm states - relates this back to quantizing A_μ in QED where we need to use gauge invariance to kill negative norm states that show up when you quantize A_0 .

It's going to be convenient to write our anomaly in a number of ways. First, recall the final thing Cameron derived:

$$\partial_\mu \Gamma_{\alpha\beta\gamma}^{\mu\nu\rho} \propto D_{\alpha\beta\gamma} = \frac{1}{2} \text{tr}[\{T_\alpha, T_\beta\}T_\gamma]$$

We can also extend our calculations from before to get more general expressions for the adjustment to our path integral. For example, under a generic:

$$\Psi_L \rightarrow e^{ig\alpha_a(x)T_a^R} \Psi_L \quad A_\mu \rightarrow \Omega A_\mu \Omega^{-1} + \frac{i}{g} \Omega \partial_\mu \Omega^{-1} \quad \Omega = e^{i\alpha_a(x)T_a}$$

$$\mathcal{Z}[A] \rightarrow \exp \left(\frac{ig^3 A(R)}{32\pi^2} \int d^4x \alpha^a(x) \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^b F_{\rho\sigma}^c \text{tr} [T_a \{T_b, T_c\}] \right) \mathcal{Z}[A]$$

Here $A(R) = 1$ if our fermions are in the fundamental representation, and if not:

$$\text{tr} T_a^R \{T_b^R, T_c^R\} = A(R) \text{tr} T_a \{T_b, T_c\}$$

Finally, we can derive a related expression for diagrams that look like the chiral anomaly:

$$\mathcal{Z}[A] \rightarrow \exp \left(U(1) \text{ triangle diagram} + \frac{ig^2 q I(R)}{32\pi^2} \int d^4x \alpha(x) \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}^a F_{\rho\sigma}^b \text{tr} [\{T_b, T_c\}] \right) \mathcal{Z}[A]$$

$$I(R) = 2 \text{tr} (T_\alpha^R)^2$$

So we have many ways to prove that an anomaly cancels. We can show that $D_{\alpha\beta\gamma} = 0$ for every combination of generators, we can show that $A(R) = 0$ for every rep, we can show that q_i^3 sums to 0, or we can show that $qI(R)$ sums to 0.

We can use these to show that only a few gauge groups actually have anomalies. Let's examine $D_{\alpha\beta\gamma}$ for a conjugate representation:

$$\bar{T}_\alpha = -T_\alpha^* = -T_\alpha^T (= U(T_\alpha)U^{-1})$$

If we plug this into our expression for

$$D_{\alpha\beta\gamma} = \frac{1}{2} \text{tr} [\{\bar{T}_\alpha, \bar{T}_\beta\} \bar{T}_\gamma] = -\frac{1}{2} \text{tr} [\{(T_\alpha)^T, (T_\beta)^T\} (T_\gamma)^T] = -\frac{1}{2} \text{tr} [\{T_\alpha, T_\beta\} T_\gamma] = (-D_{\alpha\beta\gamma} = 0)$$

We have proven two things. First: $A(\bar{R}) = -A(R)$. So our anomaly cancels if our left-handed fermions come in $R, \bar{R}$ pairs - OR we can just have an equal number of left/right handed fermions!

Second, what if we stated that our rep was equivalent to its conjugate? Then $D_{\alpha\beta\gamma} = 0$! This condition is equivalent to the gauge group having either a real or pseudoreal representation, which eliminates lots of groups! We are left with $SU(N \geq 3)$, $SO(4N+2)$, and E_6 . I'm not going to prove this, but it turns out that $D_{\alpha\beta\gamma} = 0$ for E_6 and $SO(4N+2 > 10)$. So we really just have

$$SU(N \geq 3)$$

(as $SO(6)$ has the same Lie algebra as $SU(4)$, and $SO(2)$ is just $U(1)$). (Un)luckily for us, these are the groups that show up extremely often in particle physics!

2.1 $SU(2)$ Anomaly

You'll notice that $SU(2)$ is missing. It turns out there is an anomaly in $SU(2)$, but it is sneakier! All representations of $SU(2)$ are either real or pseudoreal. However, it turns out that if we try to couple an $SU(2)$ gauge theory to an odd number of massless, left-handed fermions in $SU(2)$ doublets, the result is mathematically inconsistent.

I won't show this, but schematically the reason is that when you calculate the path integral, usually you end up with $\det(i\cancel{D})$ after integrating out the fermions, but if you have an odd number, you end up with $\sqrt{i\cancel{D}}$, which has a $\pm$ ambiguity. This would be fine if choosing $\pm$ was gauge invariant, but it turns out that it isn't, as there are non-trivial gauge transformations in $SU(2)$ that map $\sqrt{\det(i\cancel{D})}$ to $-\sqrt{\det(i\cancel{D})}$. This is the "Witten anomaly."

3 Standard Model

All of this brings us to the Standard Model, which has 3 problematic gauge groups!

Fermion	$SU(3)$ rep	$SU(2)$ rep	$U(1)$ rep
$l_L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$	1	2	-3
$q_L = \begin{pmatrix} u \\ d \end{pmatrix}_L$	3	2	1
u_R^*	$\bar{3}$	1	-4
d_R^*	$\bar{3}$	1	2
e_R^*	1	1	6

We want to show that the anomaly contributions from every possible triangle diagram cancel. Luckily, there are a few we can get rid of immediately, noting that if T_α is a generator of a different group than T_β and T_γ , $D_{\alpha\beta\gamma} \propto \text{tr}(T_\alpha)$. So we cannot have an anomaly with just one $SU(2)$ or $SU(3)$ vertex, as the trace of a non-abelian generator is 0 - there need to be either 0, 2, or 3.

Let's consider the single gauge diagrams first.

If we have three $SU(3)$ vertices, we note that there are an equal number of left-handed and right handed quarks, so all of our $A(R)$'s cancel!

If we have three $SU(2)$ vertices, we once again have cancellation trivially. Perturbatively, we know that $SU(2)$ has only real/pseudoreal reps, which means $D_{\alpha\beta\gamma} = 0$. Non-perturbatively, we see that we have an even number of fermion doublets, so we avoid the Witten anomaly.

Finally, if we have three $U(1)$ vertices, we need to add up q^3 for all fermions, weighted by the dimension of the other representations:

$$2(-3)^3 + 6(1)^3 + 3(-4)^3 + 3(2)^3 + 1(6)^3 = -54 + 6 - 192 + 24 + 216 = 0!$$

Great. We move onto the mixed anomalies, which both include one $U(1)$ leg. We sum over each representation individually, since the $I(R)$ factor is the same for each rep and can be factored out (3 and $\bar{3}$ are the same?). First, two $SU(3)$ legs:

$$2(1) + 1(-4) + 1(2) = 0$$

$$2(-3) + 1(6) = 0$$

Then two $SU(2)$ legs:

$$3(1) + 1(-3) = 0$$

$$3(-4) + 3(2) + 1(6) = 0$$

There is one last thing we should check - what happens if we couple the Standard Model to gravity? Only a problem for Abelian gauge groups - all fermions couple to gravity the same way, so if there is just a single non-Abelian leg, the diagram vanishes, as before. We also need two graviton legs (why?), so in the case of the SM, the only gauge group we care about is therefore $U(1)$.

This means we need all the $U(1)$ charges to add up to zero. We see that they do:

$$2(-3) + 6(1) + 3(-4) + 3(2) + 6 = -6 + 6 - 12 + 6 + 6 = 0!$$

So we have no issues coupling to gravity either!

(only remaining problem - what about three gravitons? doesn't show up in 3+1 dimensions, but does in other dimensions - 1+1, 9+1, and 26! worrisome for string theorists+condensed matter theorists)

4 Model Building

Could we have assigned our $U(1)$ charges in a different way and still seen this anomaly cancellation? Turns out this is very difficult! Let's keep the same $SU(2)$ and $SU(3)$ representations, but allow the $U(1)$ charges to be arbitrary:

Fermion	$SU(3)$ rep	$SU(2)$ rep	$U(1)$ rep
$l_L = \begin{pmatrix} \nu_e \\ e \end{pmatrix}_L$	1	2	l
$q_L = \begin{pmatrix} u \\ d \end{pmatrix}_L$	3	2	q
u_R^*	$\bar{3}$	1	$-u$
d_R^*	$\bar{3}$	1	$-d$
e_R^*	1	1	$-x$

We play the same game as before, and find that for two $SU(2)$ legs and two $SU(3)$ legs, and all $U(1)$ legs, respectively:

$$3q + l = 0 \quad 2q - u - d = 0$$

$$6q^3 + 2l^3 - 3u^3 - 3d^3 - x^3 = 0$$

Then gravity:

$$2l + 6q - 3u - 3d - x = 0$$

This set of equations only has two solutions!!

$$q = l = x = 0, u = -d$$

$$x = 2l = -3(u + d) = -6q \quad u - d = \pm 6q$$

First one is boring and not chiral, the second one is what we see in nature! Note further - the ratios of the hypercharges MUST be rational!

We can also go the other way. Let's assume that all of our charges are integer quantized. Then:

$$u + d \text{ even} \implies u - d = 2y \text{ even}$$

$$\begin{aligned} 6q^3 + 2l^3 - 3u^3 - 3d^3 - x^3 = 0 &= 6q^3 + 2(-3q)^3 - 3(u^3 + d^3) - x^3 \\ \implies x^3 + 48q^3 + 3(u^3 + d^3) &= 0 \end{aligned}$$

$$(u^3 + d^3) = (u + d)(u^2 - ud + d^2) = 2q((u - d)^2 + ud) = 2q(4y^2 - ud)$$

Oops, I didn't have time before my talk to finish this math up. But the result is:

$$\implies x^3 + 18qy^2 + 54q^3 = 0$$

Since this is homogeneous, we can set $q = 1$ and solve an easier equation, then rescale everything after the fact. Solving $x^3 + 18y^2 + 54 = 0$ involves setting:

$$x = -\frac{6}{v + w} \quad y = \frac{3(v - w)}{v + w} \implies v^3 + w^3 = 1$$

So if we can solve for rational v, w , we have solved this equation. But Fermat's Last Theorem says we cannot! We can ONLY have $v = 0, w = 1$ or vice versa:

$$x = -6 \quad 2y = (u - d) = \pm 6, \text{ etc}$$

Also - nowhere here did we use the gravitational side of things, it was just automatically satisfied. This is somewhat rare!

5 Beyond the Standard Model

In terms of building models, this anomaly cancellation becomes important. Firstly, we can consider embedding the standard model in $SO(10)$, which we proved before has only anomaly free representations! So if a grand unified theory of the universe lives in $SO(10)$, it makes sense that the Standard model we observe has no anomalies. Weinberg says this, but Tong disagrees, given our calculation above...

There is also a more bottom up approach. Say you wanted to build up a $U(1)$ gauge theory from the ground with N left-handed fermions.

If $N = 2$, they have to have opposite charge. If $N = 3$, Fermat's Last Theorem again tells us that this is impossible!!

If $N = 4$, we have two options - 3 positive and 1 negative or 2 and 2. In the former case, we need:

$$x^3 + y^3 + z^3 = a^3$$

Lots of solutions - 3, 4, 5, 6! In the latter:

$$x^3 + y^3 = z^3 + a^3$$

$$1^3 + 12^3 = 9^3 + 10^3$$

Ramanujan + Hardy story!

BUT Gravity breaks all of this!! Means you need far more fermions - at least 5. Standard model gives us 15.